CIE AS PHYSICS (9702) • QUESTION PAPER

10.2 Kirchhoff's Laws

Syllabus Section: 10. D.C. Circuits **Paper Type:** Structured Questions (Paper 2) **Total:** 7 Questions (91 Marks)

Question 1 Easy [11 marks]

Part (a) [3 marks]

Three resistors of resistances R_1, R_2 and R_3 are connected as shown in Fig. 1.1.

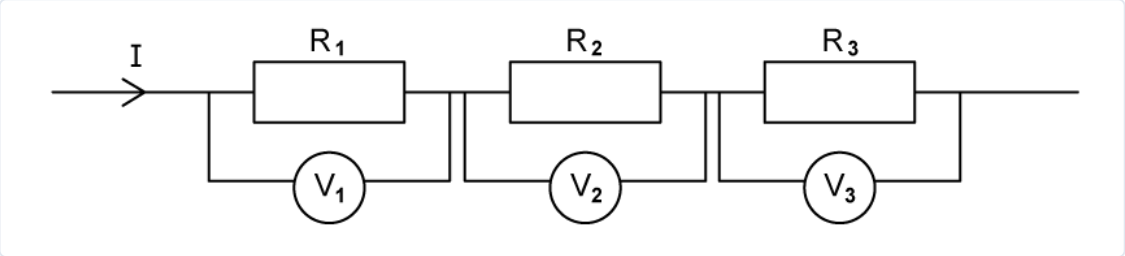


Fig 1.1

The potential difference across the resistors are V_1, V_2 and V_3. The current in the combination of resistors is I.

Show that the total resistance R of the combination is given by the equation.

$$R = R_1 + R_2 + R_3$$

Part (b) [3 marks]

Three resistors are placed in a circuit with a cell of 10 V and negligible internal resistance and two switches S_1 and S_2. This is shown in Fig 1.1.

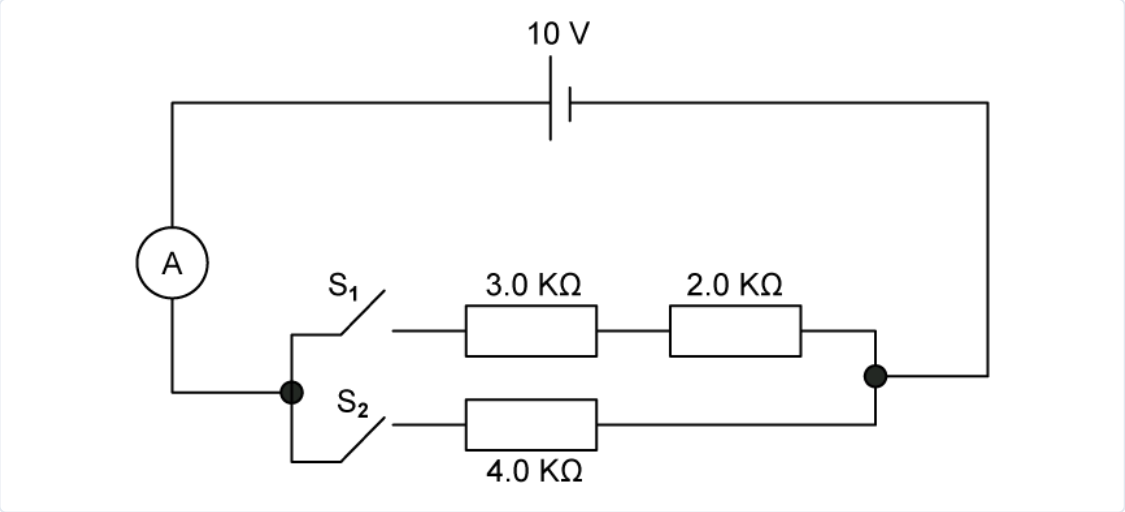


Fig 1.1

Complete table 1.1.

Table 1.1

Position of switches		
S_1	S_2	Total resistance / kΩ
open	closed	
closed	open	
closed	closed	

Part (c) **[2 marks]**

The cell has an e .m .f of 10 V and negligible internal resistance.

Calculate the current from the cell when

(i) switch S_1 is closed,

[1]

(ii) both S_1 and S_2 are closed.

[1]

Part (d) **[3 marks]**

Using your answers in part (c), determine the current from the cell if just switch S_2 is closed.

Question 2 Easy [12 marks]

Part (a) [2 marks]

(i) State Kirchhoff's first law.

[1]

(ii) State the quantity that is conserved by this law.

[1]

Part (b) [2 marks]

(i) State Kirchhoff's second law.

[1]

(ii) State the quantity that is conserved by this law.

[1]

Part (c) [6 marks]

Fig 1.1 shows a list of circuit components.

Earth	
Variable resistor	
Heater	
LED	
Thermistor	
Microphone	

Fig 1.1

In the right hand column, sketch the circuit symbol for the component in the left hand column of Fig 1.1.

Part (d) [2 marks]

Two cells of e.m.f. E_1 and E_2 are negligible internal resistance are connected to resistors R in a circuit shown in Fig 1.2.

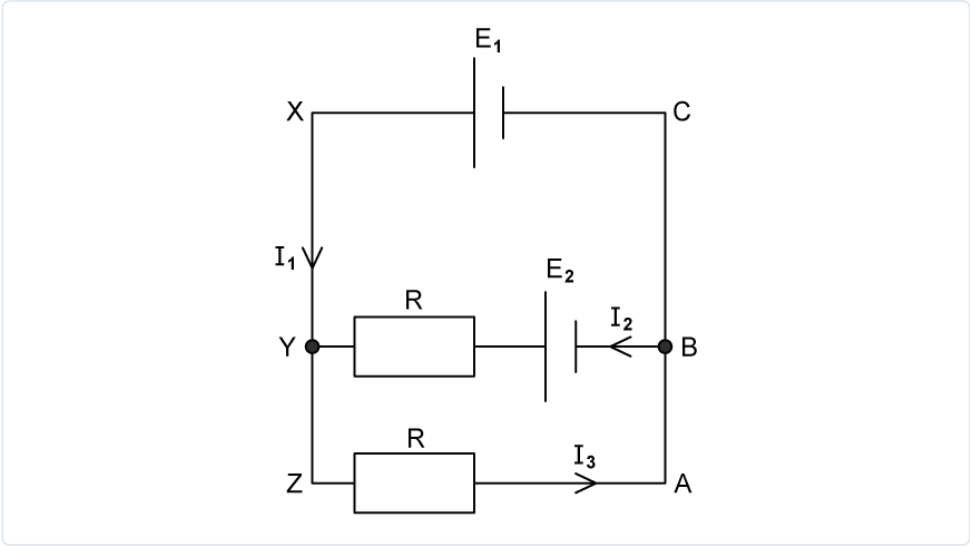


Fig 1.2

Use Kirchhoff's laws to state the relation between

(i) I_1, I_2 and I_3

[1]

(ii) E_2, R, I_2 and I_3 in loop YZABY.

[1]

Question 3 Medium [12 marks]

Part (a) [2 marks]

Three resistors of resistances R_1, R_2 and R_3 are connected as shown in Fig. 1.1.

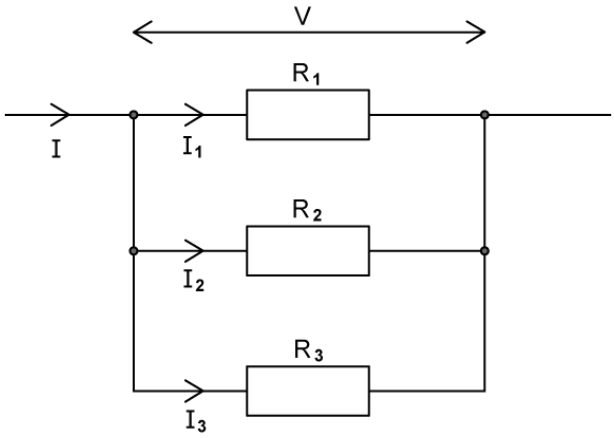


Fig. 1.1

The currents in the resistors are I_1, I_2 and I_3. The total current in the combination of resistors is I and the potential difference across the combination is V.

Show that the total resistance R of the combination is given by the equation

$$\frac{1}{R} = \frac{1}{R_1} + \frac{1}{R_2} + \frac{1}{R_3}$$

Part (b) [10 marks]

A battery of electromotive force (e.m.f.) 6.0 V and internal resistance r is connected to an external resistor of resistance 12 Ω and a thermistor X, as shown in Fig. 1.2.

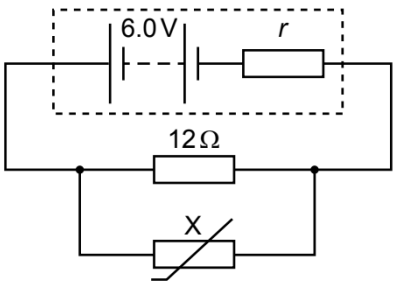


Fig. 1.2

(i) By considering energy, explain why the potential difference across the terminals of the battery is less than the e.m.f.

[1]

(ii) A charge of 2.5 kC passes through the battery.

Calculate:

- the total energy transferred by the battery

energy = ... J

- the number of electrons that pass through the battery.

number = ...

[3]

(iii) The combined resistance of the external resistor and thermistor X connected in parallel is 4.8 Ω.

Calculate the resistance of X.

resistance = ... Ω [1]

(iv) Use your answer in **(b)(iii)** to determine the ratio

$$\frac{\text{power dissipated in thermistor X}}{\text{power dissipated in 12 Ω resistor}}$$

ratio = ... [2]

(v) The temperature of thermistor X is now decreased.

State and explain the effect, if any, of this temperature change on the total power produced by the battery.

[3]

Question 4 Medium [15 marks]

Part (a) [1 mark]

A student investigates the potential difference in a circuit. The circuit is set up as shown in Fig. 2.1.

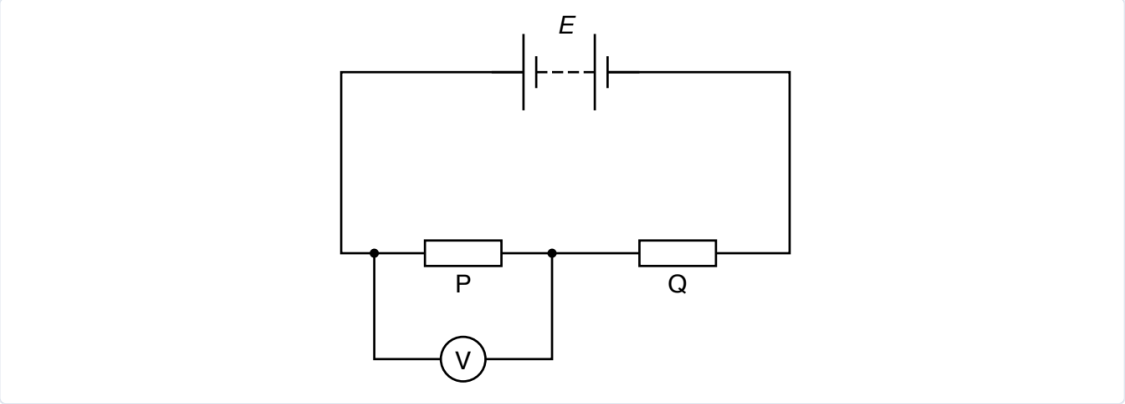


Fig. 2.1

Two resistors P and Q are connected in series to a power supply of electromotive force (e.m.f.) E and negligible internal resistance. Resistor P has resistance P.

The potential difference V across resistor P is measured. The experiment is repeated for different values of P.

It is suggested that V and P are related by the equation

$$V = \frac{P}{P+Q}E$$

where Q is the resistance of resistor Q. The value of Q is kept constant.

A graph is plotted of $\frac{1}{V}$ on the y-axis against $\frac{1}{P}$ on the x-axis.

Determine expressions for the gradient and the y-intercept.

gradient = ...

y-intercept = ...

.....

.....

Part (b) [2 marks]

Values of P, V and $\frac{1}{V}$ are given in Table 2.1.

Table 2.1

P / Ω	V / V	$\frac{1}{P}$ / 10^{-3} Ω^{-1}	$\frac{1}{V}$ / V^{-1}

250	0.66		1.52
330	0.86		1.16
470	1.15		0.870
560	1.30		0.769
680	1.49		0.671
840	1.64		0.610

Each value of *P* has an uncertainty of ±10%.

Calculate and record values of $\frac{1}{P}$ / 10^{-3} Ω^{-1} in Table 2.1.

Include the absolute uncertainties in $\frac{1}{P}$.

.....

.....

.....

.....

Part (c) **[8 marks]**

(i) Plot a graph of $\frac{1}{V}$ / V^{-1} against $\frac{1}{P}$ / 10^{-3} Ω^{-1}.

Include error bars for $\frac{1}{P}$.

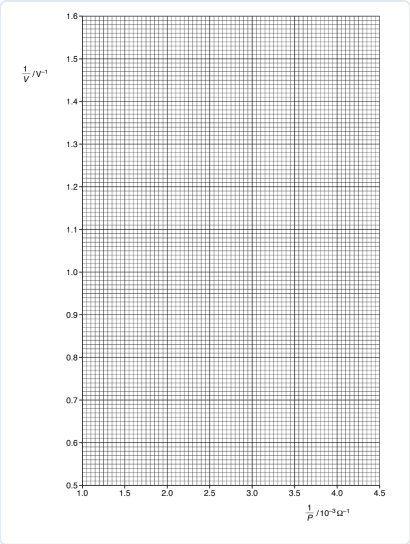
[2]

(ii) Draw the straight line of best fit and a worst acceptable straight line on your graph. Label both lines.

[2]

(iii) Determine the gradient of the line of best fit. Include the absolute uncertainty in your answer.

gradient = ... [2]



(iv) Determine the y-intercept of the line of best fit. Include the absolute uncertainty in your answer.

y-intercept = ... [2]

.....

.....

.....

.....

.....

.....

.....

.....

Part (d) **[4 marks]**

(i) Using your answers to **(a), (c)(iii)** and **(c)(iv)**, determine the values of E and Q. Include appropriate units.

E = ...

Q = ...

[2]

(ii) Determine the percentage uncertainty in *E*.

percentage uncertainty in *E* = ... % [1]

(iii) Determine the absolute uncertainty in *Q*.

absolute uncertainty in *Q* = ... [1]

.....

Question 5 Medium [12 marks]

Part (a) [4 marks]

The circuit shown in Fig. 1.1 contains a battery with an e.m.f. E and negligible internal resistance r connected to four resistors R_1, R_2, R_3 and R_4, each of resistance R.

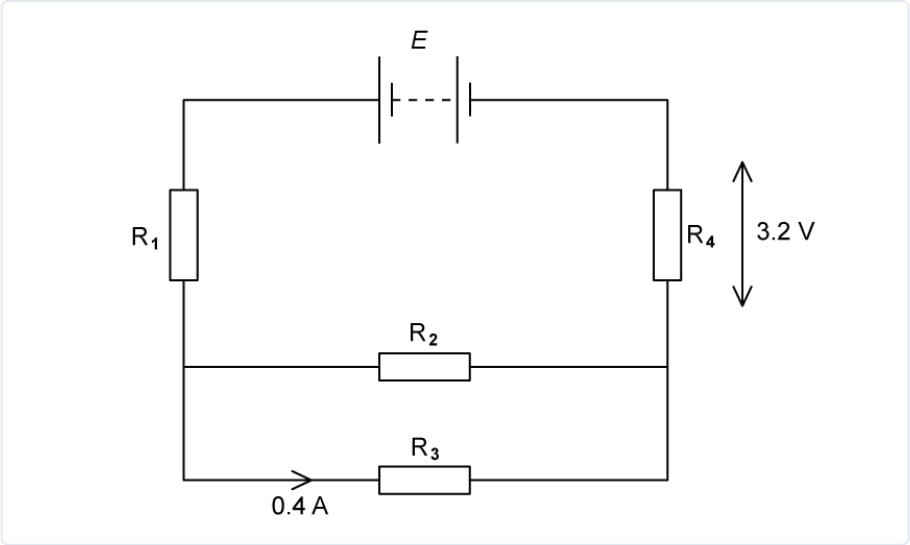


Fig. 1.1

The current in R_3 is 0.40 A and the potential difference across R_4 is 3.2 V.

Determine the resistance R of each of the fixed resistors.

Part (b) [2 marks]

Determine the e.m.f. of the battery.

Part (c) [3 marks]

A fifth resistor R_5 also with resistance R is now added in parallel to R_2 and R_3. The current through R_3 remains at 0.40 A.

Determine the total current in this circuit. Explain your answer.

Part (d) [3 marks]

The battery in the circuit is now replaced by one with the same e.m.f. but internal resistance is not negligible.

Explain whether this will change the total power produced by the battery.

Question 6 Hard [14 marks]

Part (a) [4 marks]

A circuit containing a junction P has two power supplies with e.m.f 6.0 V and 2.0 V and two resistors with resistance 30 Ω and 10 Ω is shown in Fig. 1.1.

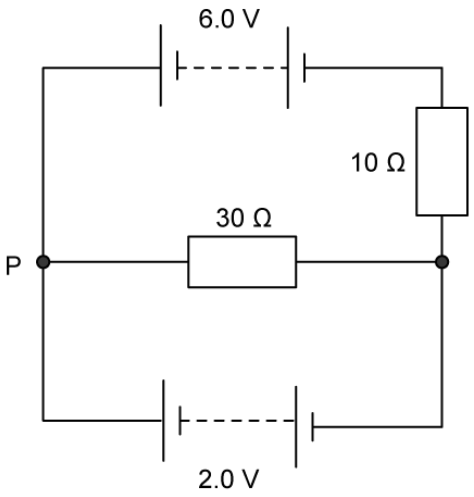


Fig. 1.1

Draw on the diagram to label the currents in each branch of the circuit and obtain an expression for the current at P.

Part (b) [4 marks]

Determine the values of the currents at P for the circuit in part a).

Part (c) [2 marks]

A combination of identical resistors each with resistance R has a total resistance of 250 Ω as shown in Fig. 1.2.

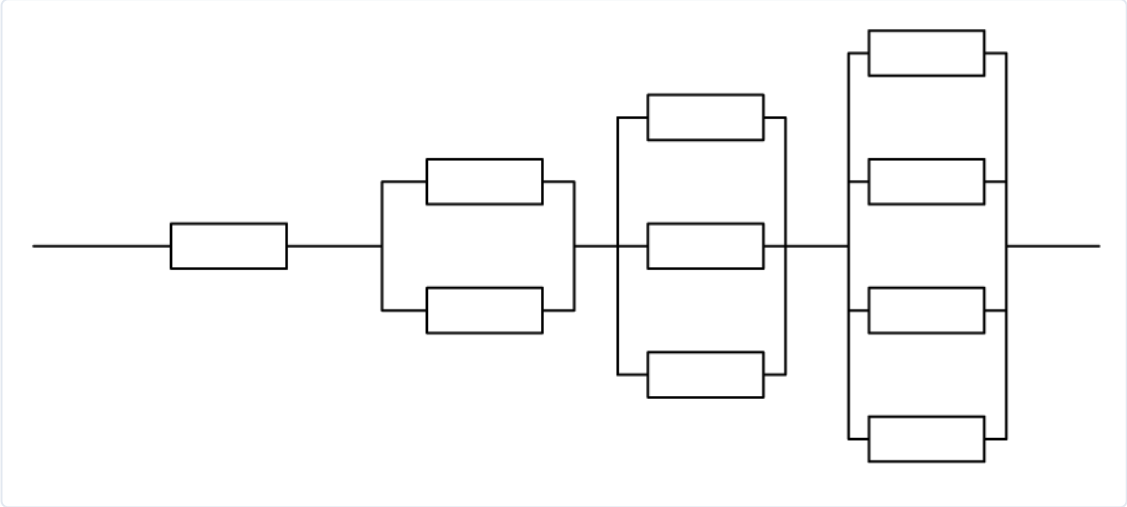


Fig. 1.2

Show, without the use of a calculator, how to find the value of resistance of each resistor and hence determine the value of R.

Part (d) [4 marks]

A student wishes to determine the resistance for three different resistors. They connect the resistors in a 'T' arrangement and change the position of the power source and the voltmeter, as shown in Fig. 1.1**,** to obtain two sets of results, as shown in Table 1.1. The power supply in Circuit 1 causes a current of 0.50 A in one branch of a circuit.

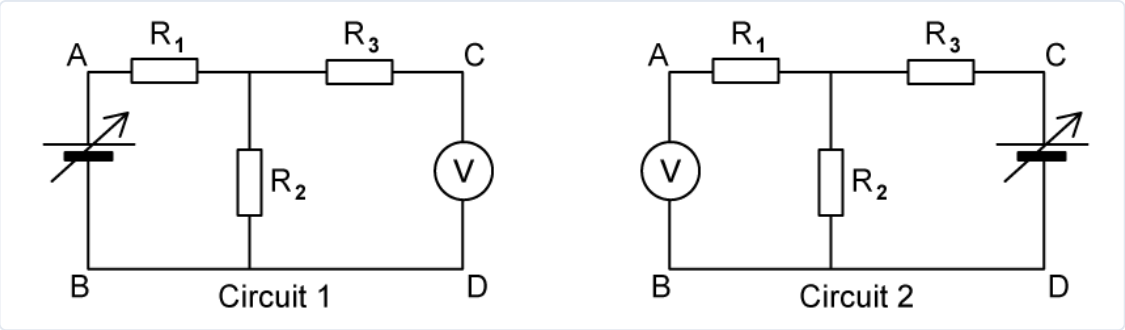


Fig. 1.1

Table 1.1

Circuit	Voltage supplied by power source /V	Reading on voltmeter /V
1	8.0	5.0
2	10	4.0

Calculate the resistance of resistors R_1 , R_2 and R_3 .

You may assume the voltmeter has an infinitely high resistance.

Question 7 Hard [15 marks]

Part (a) [4 marks]

There is current flowing from P to R in the resistor network shown. The potential differences across three of the resistors are 3.0 V, 6.0 V and 5.0 V as shown in Fig. 1.1.

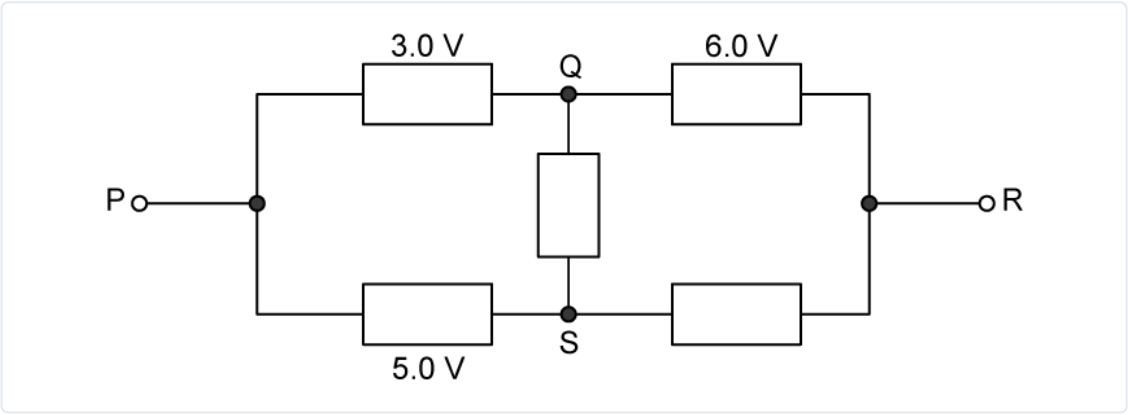


Fig. 1.1

Determine the potential difference across the other two resistors.

Part (b) [4 marks]

The current through the branch QS is 3.0 A. The total combined resistance of the resistors in the outer loop of the circuit is *r*. The resistance in the outer loop of the circuit is split according to the ratio of the magnitudes of the potential differences of the resistors on each side of the loop.

Obtain an expression, in terms of *r*, for the total resistance of all the resistors in the circuit.

Part (c) **[7 marks]**

Diagram 1.2 shows a circuit where V_1 is a battery of 0.5 V, V_2 a battery of 1 V and V_3 a battery of 2 V. Each resistor has a resistance of 0.25 Ω.

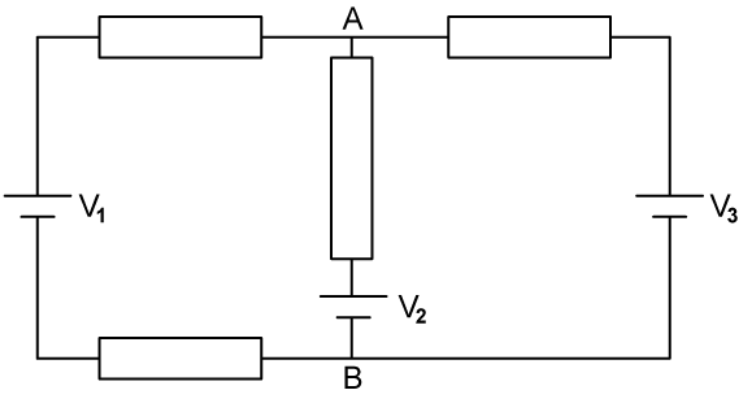


Fig. 1.1

Determine the value of each current at junction A.